specifically designed for this service. This accelerator is available only in Oracle cloud infrastructure (OCI) and optimised only for it. It always ensures that real-time data is being processed whenever you run the queries, and provides an overall better price to performance ratio. The query processing time and the performance is 5,400 times faster than that of its competition.

Figure 1 shows how HeatWave is integrated into InnoDB. As can be seen, MySQL has InnoDB and HeatWave built in. HeatWave has an Autopilot advisor that helps get recommendations, which makes queries run faster. Let us now move on to a more detailed architecture for better clarity.

As can be seen in Figure 2, the OLAP queries get optimised by the optimiser present in the MySQL database service itself. When the HeatWave cluster is enabled, queries that meet certain prerequisites are automatically offloaded from the MySQL DB system to the HeatWave cluster for accelerated execution without any user intervention. The HeatWave node gives recommendations and the query estimation time, and then executes these within a very short time compared to a regular OLAP processor. It then sends the results back to the MySQL database service.

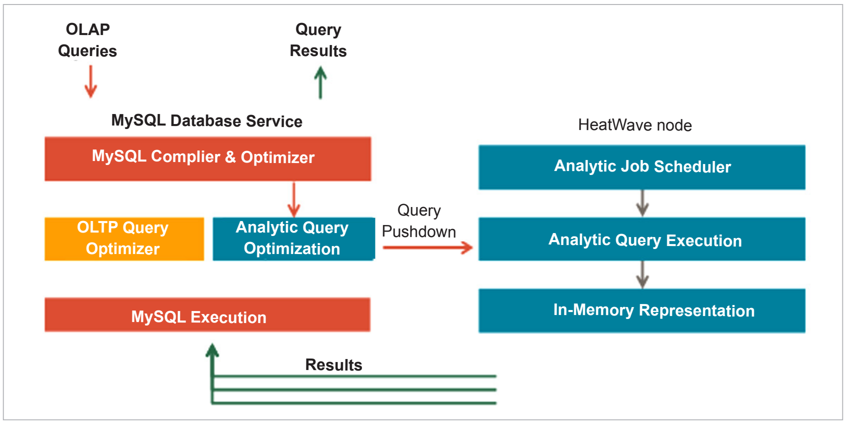

Figure 2: Detailed HeatWave architecture

How is MySQL Autopilot making a difference?

MySQL Labs has been working on Autopilot for at least a decade to enhance its machine learning (ML) capabilities. This optimiser has been increasingly made intelligent with the integration of ML algorithms leading to an excellent performance. It gives smart recommendations based on the data, and only uses 0.1 per cent of the data sample to come up with a query estimation. The query performance at scale is also high, improving the usability of the system. The system is used effectively, especially with respect to data placement or managing change propagation.

Oracle MySQL Autopilot: How it works

MySQL Autopilot helps in four major domains.

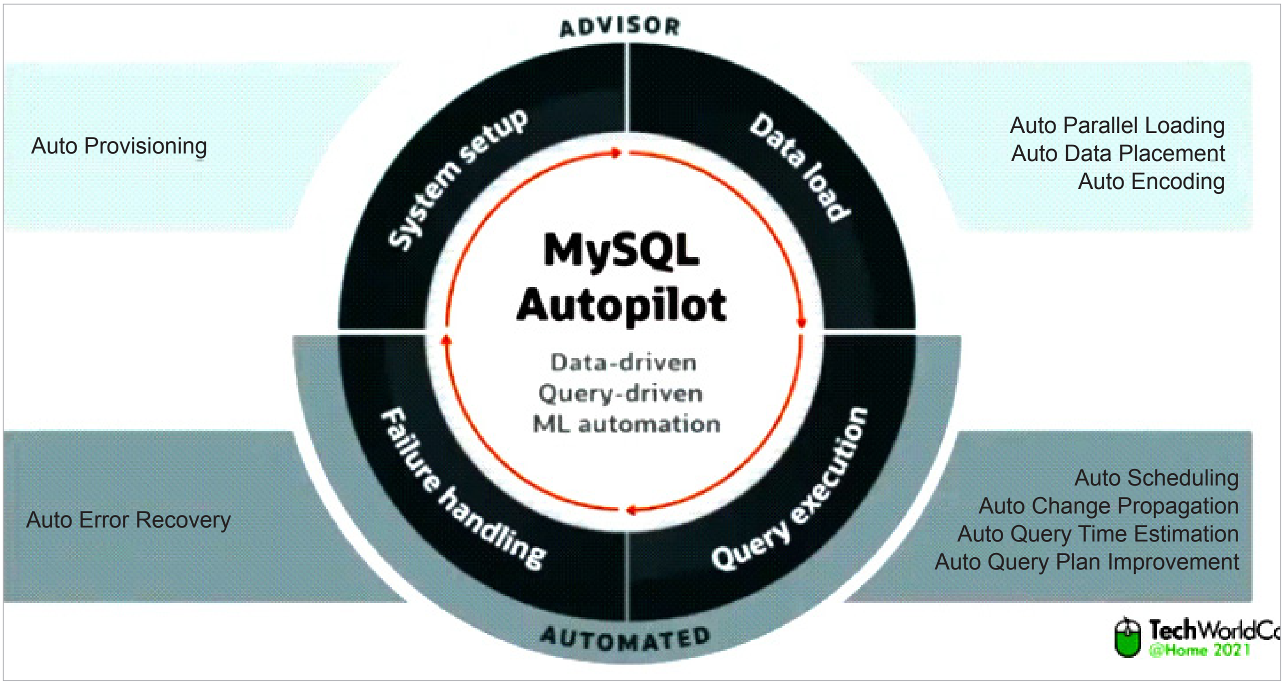

Figure 3: MySQL Autopilot